Introduction to Databases

A database is an organized collection of structured data stored electronically. A database management system (DBMS) is the software that lets users create, read, update, and delete data. Examples of relational DBMS software include PostgreSQL, MySQL, and Oracle.

Relational databases store data in tables. Each table has rows, called records, and columns, called attributes. Every table should describe one kind of thing, such as students or courses.

A primary key is a column, or a set of columns, that uniquely identifies each row in a table. A primary key value cannot be NULL and cannot be repeated. For example, student_id can be the primary key of a students table.

A foreign key is a column in one table that refers to the primary key of another table. Foreign keys create relationships between tables and enforce referential integrity, which means a row cannot refer to a record that does not exist.

Database Normalization

Normalization is the process of organizing tables to reduce data redundancy and avoid update anomalies. Redundant data wastes storage and can become inconsistent when only some copies are changed.

First Normal Form (1NF) requires that every column holds atomic values, meaning a single value per cell, and that there are no repeating groups. A column that stores a list of phone numbers violates 1NF.

Second Normal Form (2NF) requires the table to be in 1NF and every non-key attribute to depend on the whole primary key. This rule removes partial dependencies and matters for tables with composite keys.

Third Normal Form (3NF) requires the table to be in 2NF and to have no transitive dependencies. A non-key attribute must depend only on the primary key, not on another non-key attribute.

SQL Joins

A join combines rows from two or more tables based on a related column, usually a foreign key matched to a primary key.

An INNER JOIN returns only the rows that have matching values in both tables. Rows without a match are left out of the result.

A LEFT JOIN returns all rows from the left table and the matching rows from the right table. When there is no match, the right-side columns contain NULL.

A FULL OUTER JOIN returns all rows from both tables, filling NULL where there is no match. A CROSS JOIN returns every combination of rows from the two tables, known as the Cartesian product.

Transactions and Indexes

A transaction is a group of SQL operations that are treated as a single unit of work. Transactions follow the ACID properties: Atomicity, Consistency, Isolation, and Durability.

Atomicity means all operations in a transaction succeed or none of them do. Durability means that once a transaction is committed, its changes survive a system crash.

An index is a data structure that speeds up searches on a column, similar to the index of a book. Most relational databases use B-tree indexes by default. Indexes make reads faster but slow down inserts and updates, because the index must also be maintained.

A good practice is to create indexes on columns that are often used in WHERE clauses and join conditions, and to avoid indexing columns that are rarely searched.